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RESEARCH ARTICLE

Model Between Network Quality and XR User's Experience in Terms of Video Stalling for Cloud Rendering XR

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ABSTRACT At present, extended reality (XR) is a hotspot of research around the world. The cloud rendering XR based on mobile communication system provides a significant solution to reducing the demand of head-mounted display (HMD) for computing power, thus the scope of application for XR can be expanded from indoors to outdoors. However, cloud rendering XR has more demanding requirements on network quality, failure to meet these requirements leads to poor user experience. In particular, video streaming stalling is the most critical factor affecting the immersive experience, and it is closely related to network quality. Therefore, it is essential that the network is capable of predicting the user experience of stalling for precise resource scheduling and allocation. In order to reveal the relationship between network quality and stalling, a frame-level data transmission framework with two-layer mapping model is proposed in this paper. The framework uses indicators like packet loss, latency, throughput, and forward error correction (FEC) to model video frame loss in the lower layer. The upper layer then uses this information to predict stalling probability. Furthermore, simulation and experiment are implemented to evaluate the accuracy and feasibility of the proposed model in the 5th generation mobile communication technology (5G). The simulation results show that the data packet loss of less than 1.4% can ensure user to experience without stalling. The experimental results indicate that for 60 Mbps and 30 Mbps encoding rate, the data packet loss for no stalling are measured to be less than 1% and 2%.

INDEX TERMS Extended reality, user experience, network quality, video stalling.

I. INTRODUCTION

Extended Reality (XR) is defined as the integration of real and virtual environments through computer technology to create interactive virtual spaces for both humans and machines. It is the general term for virtual reality (VR), augmented reality (AR), mixed reality (MR). With the rapid development of 3D optical displays and 3D user interfaces, significant advancements have been made across various domains of XR services, revolutionary changes and innovative experiences have been brought to people's production [1] and lifestyle [2]. Popularity of XR services has seen a huge growth [3], [4], [5], [6], and cloud rendering XR is poised to become the primary form of XR services. Cloud rendering XR offloads computing and rendering tasks to cloud servers, transmits the video streams to user HMD. This approach can reduce the requirements for local rendering, storage, and processing of large data in HMD [7]. However, cloud rendering XR can be considered as a form of video streaming with extremely high network transmission requirements. Numerous studies indicate that network quality is one of the dominant factors affecting user experience [8], [9], [10]. Especially the video stalling directly impacts the user's sense of immersion and satisfaction, and it is inextricably linked to network quality [11]. Video stalling typically refers to the playback interruptions or screen freezes caused by insufficient

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bandwidth or server processing capacity limitations [12], [13], and it can lead users feeling sickness and force them to stop using the XR service [15], [16]. In cloud rendering XR, video frames are generated and rendered by servers deployed in the cloud or edge based on user actions. The latency requirements for each frame transmission to the HMD are more strict, and prolonged delay or severe data packet loss may cause the video frame playback interruptions [14].

5G new radio (NR) is the new generation mobile communication system and can provide reliable network transmission capability for XR service [17], [18], [19], [20]. With the advantages of high throughput, low latency and low loss, 5G can enable the cloud rendering XR service to gradually expand the boundaries of usage scenario from indoors to outdoors. XR users can get more immersive experience in a boundary-less virtual world by 5G. It is a crucial target for 5G to provide higher priority assurance for cloud rendering XR to ensure the experience of no stalling. However, higher priority assurance of network quality means more usage of transmission resource. Network is supposed to have the ability to determine whether the user feels stalling according to the network quality indicators, and adjust the network transmission strategy to ensure better user experience in the balance of transmission resources. Currently, the network quality indicators and user experience indicators are respectively measured in two separate dimensions. This separation results in inflexible, imprecise, and unreal-time scheduling of the transmission resources in network, which may affect the experience of other users in the cell.

In this paper, a frame-level data transmission framework with two-layer mapping model is proposed to reveal the relationship between the network quality and the stalling, simulation and experiment are carried out to evaluate the accuracy and feasibility of the proposed model in 5G. The proposed model has the following three contributions:

- In the lower-layer model, the mapping relationship between network quality indicators and video frame loss is comprehensively addressed. In the upper-layer model, an innovative probability distribution model is proposed to investigate the impact of video frame loss on stalling. Through the above two-layer mapping model, the probability of stalling can be predicted based on varying network quality indicators.
- The probability of stalling under various packet loss conditions is simulated with different frame per second (FPS), coding rate and re-transmission latency based on the proposed model. The results show that data packet loss is supposed to be less than approximately 1.4% for experience of no stalling.
- The experiment in 5G network is implemented to evaluate the accuracy and feasibility of the proposed model. The results show that the proposed model performs extremely well, and it can serve as a foundational model for the precise allocation of resources in the network transmission in the future.

The remainder of this paper is organized as follows. In Section II, the related work in the last decade is briefly discussed. In Section III, a frame-level data transmission framework with two-layer mapping model is presented, including theoretical analysis and mathematical modeling. In Section IV, the simulation is implemented to verify the feasibility of the proposed model based on Section II, and experimental test is also carried out to testify accuracy by a real cloud rendering VR game in a 5G test network with commercial configuration. Finally, the conclusion and future work are discussed in Section V.

II. RELATED WORK

In the last decade, numerous studies have been reported to investigate theoretical models and experimental analyses of the relationship between network quality and video stalling. The dynamics of the playback buffer is modeled by the G/G/1 queue theory to predict video interruption probability for video streaming [21], the closed-form expressions of the video stalling are discussed in terms of the packet loss, fluency of video playback and start-up delay. Furthermore, based on a given video length, a queuing theoretical method is proposed to predict the video frozen and stalling probability caused by random packet delay [22]. The Markov modulated fluid model is introduced to compute the probability of stalling and the start-up delay in the network for video streaming applications [23]. The above studies focus on the analytical model in traditional video playback, however, the characteristics of the video frame have not been comprehensively considered. Due to the fast fading and slow fading of shadowing in wireless channel, the performance of mobile network transmission is rapidly changing within each video frame transmission time [24], [25]. The fine-grained network quality is supposed to be evaluated based on channel state information of each video frame, especially in cloud rendering XR. The traditional buffer model cannot effectively address this problem. Besides, the mean opinion score (MOS) [26], [27], [28] is proposed to evaluate the quality of experience (QoE) for traditional video services in mobile communication system. Subjective user experiences are mapped to objective scores, which are directly influenced by network quality. A lot of experiments have been carried out to obtain theoretical mapping relationship and improve accuracy. However, the subjective experience and network quality of cloud rendered XR differ from those of traditional video services and need to be further expanded to cloud-rendered XR.

With the increasing popularity of XR services, a variety of researches have been reported to address the issue of the video stalling for XR [29], [30], [31], [32], [33], [34], [35]. Subjective experiments investigating the impact of video frame loss on stalling are conducted [29], the results indicate that isolated stalling of up to 4 frames at 90 FPS do not lead to perceived stalling by users. However, the study is only focused on subjective tests without mathematical modeling

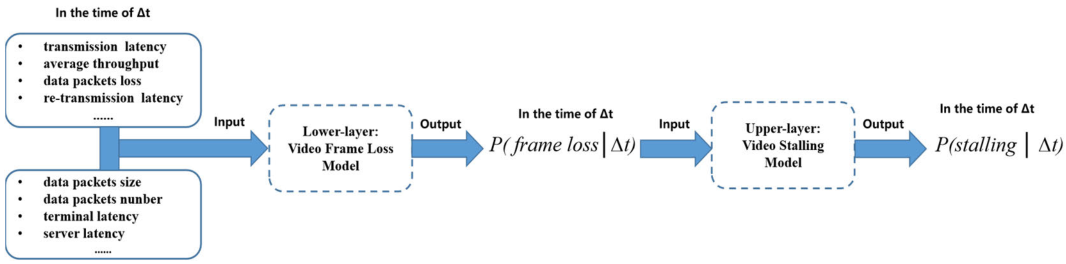

FIGURE 1. Process of the frame-level data transmission framework.

and analysis, which can not be flexibly used in real-time network with various services. For 360-degree VR video, the mapping relationship between network bit rate and various stalling events is investigated [30], and a Bayesian statistical method for measuring the QoE for 360-degree videos is proposed, which analyzes the influence of each factor on QoE [31]. The results show that a strong dependency between network bit rate and stalling is confirmed. The human perceptual tolerance to the velocity mismatch is modeled through a psycho-physiological experiment for cloud rendering VR [32], the result shows that the limited capacity and bandwidth of network transmission can cause the users to feel the stalling and suffer from sickness and dizziness. The impact of motion to photon (MTP) latency and motion-to-update (MTU) latency on a mobile VR system used in a mobile edge cloud (MEC) is analyzed [33]. The extra latency introduced by MEC may persist in affecting user experience of stalling and distorting artifact. However, the above studies only focus on individual network quality indicators, such as transmission latency, bandwidth, bit rate. Other significant indicators like the packet loss and re-transmission latency are also crucial for evaluating network quality, a poor performance of those indicators can lead to the video stalling as well. The proposed models may not adequately present comprehensive network quality. In addition, many papers have been reported that the MOS can be further updated suitable for XR service based on the QoE model [34], [35]. A QoE model for 360-degree VR video is proposed based on subjective quality evaluation, with mathematical analysis of factors such as stalling [34]. A QoE model named XR quality index (XQI) is proposed to evaluate network quality for XR services with potential applications in optimizing network transmission [35]. These QoE models generate MOS scores based on network quality to reflect user experience. However, the detailed mapping relationships of each indicator within the network are not considered in the QoE models, and a final score cannot effectively support fine-grained scheduling and resource allocation in the network.

In general, the theoretical model reported in the studies of traditional video streaming may not effectively address the characteristics of XR service, the reported model would not be applied suitably for analysis and modeling on cloud rendering XR. The methods and experimental results reported for XR services may not be directly applicable for networks to meticulously schedule and allocate transmission resources, which is necessary to ensure immersive experience for user. In this research, an frame-level data transmission framework with two-layer mapping model is proposed as shown in Figure 1. The network quality indicators are simultaneously and completely considered in the lower-layer model, and the mapping relation is addressed with video frame loss. That is not comprehensively analyzed and considered in previous studies. In the upper-layer model, video frame loss is mathematically modeled in relation to user experience of stalling using an innovative probability distribution model. Through the above two-layer mapping model, the probability of stalling is theoretically calculated based on varying network quality indicators. The data packet loss is one of the most important indicators for network quality, and simulations and experiments of data packet loss are emphatically discussed under the various conditions of re-transmission latency, FPS and encoding rate. The results show that the predicted probability of the proposed model is accurate and feasible.

III. ANALYTICAL FRAMEWORK

In this section, we theoretically establish a frame-level data transmission framework suitable for cloud rendering XR with two-layer mapping model. The lower-layer model can be considered as a video frame loss model, while the upper-layer can be functioned as a video stalling model.

A. VIDEO FRAME LOSS MODEL

As shown in Figure 2, the data transmission process of cloud rendering XR can be divided into nine parts, which are user motion capture, uplink of the wireless access network (RAN), uplink of 5G core network (5GC), cloud server rendering, frame coding, downlink of 5GC, downlink of RAN, terminal decoding, and on-screen display. When the user's HMD captures an action command, HMD processes the command and transmits command into the XR cloud rendering server via the network uplink. The XR cloud rendering server receives the command and generates the video frames based on the action command, encodes the video frames into several data packets. The data packets are transmitted as video streaming to the HMD by the network downlink. After decoding in the HMD, the video frames are displayed on the screen for user.

The successful or loss of data transmission for a video frame mainly depends on whether the MTP latency meets the demands for the refresh rate of the HMD. The MTP latency

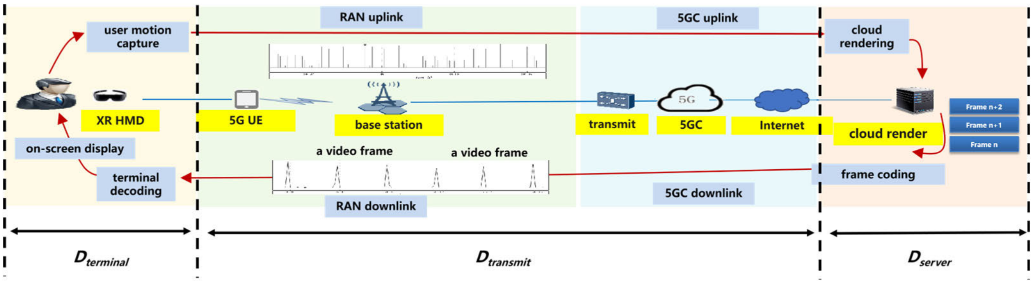

FIGURE 2. **Process of frame transmission in XR video streaming.**

of a video frame transmitted by network can be written as

$$D_{MTP} = D_{terminal} + D_{transmit} + D_{server} \tag{1}$$

where $D_{terminal}$ is denoted as terminal processing latency, which consists of user action capture latency, HMD decoding latency and on-screen display latency. D_{server} is defined as the rendering latency and encoding latency in server. $D_{transmit}$ is denoted as the sum of transmission latency, it can be further expanded as

$$\begin{aligned} D_{transmit} = {} & T_{transmit} + P(Packet\ loss|\Delta t) \\ & \cdot \mu \cdot (1 - P(Fec|\Delta t)) \cdot (T_{HARQ} + T_{retransmit}) \end{aligned} \tag{2}$$

where $T_{transmit}$ is denoted as the transmission latency in a video frame, it can be expressed as

$$T_{transmit} = \frac{\mu \cdot S_{bit}}{V_{transmit}} \tag{3}$$

where μ is the amount of data packets generated by the cloud server encoding a video frame, and S_{bit} is the average size of data packets. $V_{transmit}$ is the average data throughput of network transmission. $P(Packet\ loss|\Delta t)$ is defined as the probability of data packet loss in the network transmission at the time of Δt. Due to the fast fading and slow fading of shadowing in wireless channel, the communication system is highly susceptible to transmission errors or data packet loss in wireless transmission [27], [28]. When the data packets are loss or error after terminal decoding, the terminal can send hybrid automatic repeat request (HARQ) to the base station, and the base station enables the re-transmission of the data packets based on the HARQ. T_{HARQ} is denoted as the latency of HARQ processing, and $T_{retransmit}$ is defined as the latency of the data re-transmission in sum of network. To reduce the data packet loss and the amount of re-transmission, FEC is commonly used to correct the previous item by adding redundant data in the coding process. $P(Fec|\Delta t)$ is defined as the probability of successful error correction in the time of Δt.

Theoretically, the MTP latency for a video frame should be less than the refresh interval for each frame displayed on the HMD, which can be expressed as

$$P(Frame\ loss|\Delta t) = \Pr(D_{MTP} \leq \frac{1}{FPS}|\Delta t) \tag{4}$$

where the FPS is the frame per second of the HMD, $P(Frame\ loss\ |\Delta t)$ is defined as the probability of frame loss in the network transmission at the time of Δt. Usually, the FPS for typical VR applications ranges from 60 to 90. The theoretical requirement for network transmission MTP latency is calculated to be 11.1 to 16.6 ms in a video frame, the network transmission performance is challenging to meet the requirements. Numerous methods are proposed to reduce the limit of the latency, such as the typical secondary rendering algorithms of the asynchronous time warp (ATW) [36], [37] and the asynchronous space warp (ASW) [38], [39]. Buffer time can exist during each video frame transmission. $P(Frame\ loss|\Delta t)$ can be further rewritten as

$$\begin{aligned} P(Frame\ loss|\Delta \mathrm{t}) = {} & Pr(D_{MTP} \\ & \leq \frac{1}{FPS} + T_{buffer} + T_{ATW,ASW}|\Delta \mathrm{t}) \end{aligned} \tag{5}$$

where T_{buffer} is the time of data buffer, and $T_{ATW,ASW}$ is the time buffer caused by ATW and ASW. In order to transmit a video frame successfully, the the maximum number of data packet loss n_f can be written in (6), as shown at the bottom of the page, where the $\lfloor\ \rceil$ is denoted as the integer down.

B. VIDEO STALLING MODEL

In the above video frame loss model, video frame loss can be calculated from measurable network indicators such as latency, throughput, data packet loss and others. In this model, we further analyze the relationship between video frame loss and video stalling.

In general, user would not feel video stalling and frozen when the amount of frame loss is lower than a threshold [23]. For example, 24 FPS is an important standard for users to watch video smoothly in the field of film

$$n_f = \left\lfloor \frac{\frac{1}{FPS} + T_{buffer} + T_{ATW,ASW} - \frac{\mu \cdot S_{bit}}{V_{transmit}} - D_{terminal} - D_{server}}{(T_{HARQ} + T_{retransmit})} \right\rfloor \tag{6}$$

production [40], [41], the transmission interval between each video frame is supposed to be less than 42.2 ms. In this research, the video stalling in cloud rendering XR can be defined that the video frame cannot be transmitted to the HMD for display within a specified time threshold. Thus, for the cloud rendering XR, video frames are transmitted within Δt time, the minimum number of frames continuous loss n_j can be written as

$$\mathrm{n}_j = \left\lfloor \frac{FPS}{F_H} \right\rfloor \tag{7}$$

Where the F_H is the threshold of FPS for the smooth video display. According to the (1), (2), and (5), the total number of data packet loss n_t in Δt time, can be expressed as

$$n_t = \lfloor \mu \cdot FPS \cdot \Delta t \cdot P(Packet\ loss|\Delta t) \cdot (1 - P(Fec|\Delta t)) \rfloor \tag{8}$$

n_L is defined as the minimum number of data packet loss that causes the user stalling, it can be written in (9), as shown at the bottom of the page.

If the n_j is less than 2, the probability distribution model of stalling can be described as a classical proposition of combination mathematics, the number of the way that can partition a set of x objects into y subsets, with each subset containing up to z elements. Many methods have been reported to solve it [42], [43], in this research, recurrence method is used to calculate stalling probability. The $P(stalling|\Delta t)$ is denoted as the probability of video stalling for cloud rendering XR in the Δt time, it can be expressed as

$$p(stalling|\Delta t) = \begin{cases} 0, \ if\ n_t < n_L \\ \dfrac{(FPS \cdot \Delta t - n_j + 1) \cdot \left(C_{\mu}^{n_f}\right)^{n_j}}{C_{\mu \cdot FPS \cdot \Delta t}^{n_t}}, \ if\ n_t = n_L \\ \dfrac{P_N(\mathrm{n}_t, FPS \cdot \Delta t, n_f)}{P_N(\mathrm{n}_t, FPS \cdot \Delta t, u)}, \ if\ n_t > n_L \end{cases} \tag{10}$$

where the C is represented as the binomial coefficient, it can be defined as

$$C_{\mathrm{n}}^{k} = \frac{n!}{k!(n-k)!} \tag{11}$$

When the n_t is less than n_L, the network transmission performance can meet the requirements for smooth video display, and the video frame loss caused by data packet loss can not lead to the video stalling and display frozen. If the n_t is equal to n_L, the number of data packet loss is just meet the threshold of stalling, it can be calculated by the principle of inclusion-exclusion. When the n_t is more than n_L, the $P_N(x,y,z)$ can be denoted as the number of the way to partition the set of x data packet loss in y video frame, with each video frame only containing up to z data packet loss, the recurrence relation of $P_N(x,y,z)$ can be expressed as

$$P_N(x, y, z) = \sum_{0 \le i \le yzi \le n} C_y^i \cdot P_N(x - zi, y - i, z - 1) \tag{12}$$

the initial conditions can be expressed as

$$\begin{cases} P_N(x, y, 1) = C_y^x, \ if\ x \le y \\ P_N(x, y, 1) = 0, \ if\ x > y \end{cases} \tag{13}$$

If the n_j is more than 2, the probability model of stalling becomes more complex, which adds a limit in the original condition. It is the the number of the way that can partition a set of x objects into y subsets, with each subset containing up to z elements and no more than m subsets are continuous in serial. The $P(stalling|\Delta t)$ can be expressed as

$$p(stalling|\Delta t) = \begin{cases} 0, \ if\ n_t \ < n_L \\ \dfrac{(FPS \cdot \Delta t - n_j + 1) \cdot \left(C_{\mu}^{n_f}\right)^{n_j}}{C_{\mu \cdot FPS \cdot \Delta t}^{n_t}}, \ if\ n_t = n_L \\ \dfrac{P_{\mathrm{rm}}(n_t, FPS \cdot \Delta t, u, n_j)}{P_N(\mathrm{n}_t, FPS \cdot \Delta t, u)}, \ if\ n_t > n_L \end{cases} \tag{14}$$

When the n_t is less or equal than n_L, $P(stalling|\Delta t)$ is same as that in (10). When the n_t is more than n_L, the $P_{rm}(x,y,z,m)$ can be denoted as the number of the way to partition the set of x data packet loss in y video frame, with each video frame only containing up to z data packet loss, which is no more than m subsets are continuous in serial, the recurrence relation can be written as

$$\begin{aligned} P_{rm}(x, y, z, m) &= P_{rm}(i_1) + P_{rm}(i_2) \cdots P_{rm}(i_k) \\ &= \sum_{j=0}^{z} P_{rm}(x - j, y - 1, z, m) \\ &\quad + \sum\nolimits_{r=z}^{\mu} S(r - z, 1) \sum\nolimits_{j=0}^{z} P_{rm}(x - j - r, y - 2, z, m) \\ &\quad + \sum\nolimits_{r=2 \cdot z}^{2 \cdot \mu} S(r - 2 \cdot z, 2) \\ &\quad \sum\nolimits_{j=0}^{z} P_{rm}(x - j - r, y - 3, z, m) \cdots \\ &= \sum\nolimits_{k=0}^{m} \sum\nolimits_{r=k \cdot z}^{k \cdot \mu} S(r - k \cdot z, k) \\ &\quad \sum\nolimits_{j=0}^{z} P_{rm}(x - j - r, y - k - 1, z, m) \end{aligned} \tag{15}$$

where the $P_{rm}(i_n)$ is the number of the way that the data packet loss of n video frame is more than z, and data packet of the

$$\begin{aligned} n_L &= n_f \cdot n_j \\ &= \left\lfloor \frac{\frac{1}{FPS} + T_{buffer} + T_{ATW,ASW} - \frac{\mu \cdot S_{bit}}{V_{transmit}} - D_{ter\min al} - D_{service}}{(T_{HARQ} + T_{retransmit})} \right\rfloor \cdot n_j \end{aligned} \tag{9}$$

next video frame is less than z. $S(j,r)$ is the Stirling numbers of the second kind [43], which is described as the number of the way to partition a set of j objects into r subsets. The initial conditions can be expressed as

$$\begin{cases} P_{rm}(z, y, z, m) = y^z \\ P_{rm}(x, m, z, m) = m^x \\ P_{rm}(x, 1, z, m) = 0, \ \ if \ x > m \\ P_{rm}(x, 1, z, m) = 1, \ \ if \ x \leq m \end{cases} \tag{16}$$

IV. SIMULATION AND TEST RESULTS

In this section, simulations and experiments are carried out to evaluate the feasibility and accuracy of the proposed analytical frame-level data transmission framework with two-layer mapping model. The probability of stalling in various data packet loss is acquired by simulation with different FPS, coding rate and re-transmission latency. The accuracy of simulations result is verified in the experiments at 5G test network with a real cloud rendering VR gaming.

A. SIMULATION RESULTS

In the simulation, dynamic programming based on python is introduced to optimize the recurrence formula (10), (13), and make the programs converge quickly. As shown in Table.1, the regular parameter configuration is set to be the common value of 5G commercial network.

TABLE 1. The regular parameter configuration in simulation.

Symbol	Quantity	Symbol	Quantity
FPS	30,45,60	F_H	24
$P(Fec \mid \Delta t)$	0.8	μ	60,30
T_{HARQ} (s)	0.001	S_{bit} (KB)	1400
$D_{terminal}(s)$	0.02	$T_{retransmit}(s)$	0.0015,0.004
T_{buff} (s)	0.02	$D_{service}$ (s)	0.02
$T_{transmit}(s)$	0.02	$T_{ATW,ASW}$ (s)	0.023

As illustrated in the Figure 3, the FPS is set to be 30 and the encoding rate is set to be 60 Mbps. The latency used for each packet retransmission is set to 2.5ms. When the data packet loss are set to be 1%, 1.2%, 1.4%, 1.6%, 1.8% and 2%, the analytical results show that the stalling probabilities are approximately 0.72%, 2.68%, 7.5%, 16.87%, 32.17% and 50.35% in 60s with increasing linearly over time, respectively. Especially in the data packet loss of 1% and 1.2%, the probabilities of stalling are almost 0. When the data packet loss is changed to be higher than 3%, the probabilities of stalling increase more sharply to be nearly 100% in exponential decline over the time.

As shown in Figure 4, The latency used for each packet retransmission is set to be 5ms. Compared to the 2.5 ms retransmission latency, the reliability requirement of network transmission is more stringent. When data packet loss is set to be 0.25% and 0.5%, the probabilities of stalling are approximately 0.25% and 8.8% in 60s, which are enhanced linearly over time within the 10%. When the data packet loss

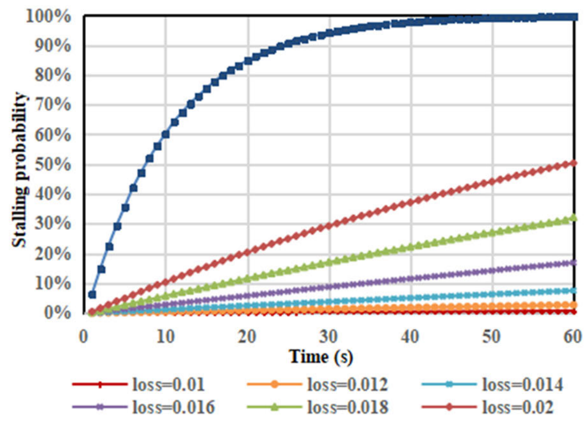

FIGURE 3. Probability of stalling in 30 FPS and 2.5 ms re-transmission latency in simulation with 60M encoding rate.

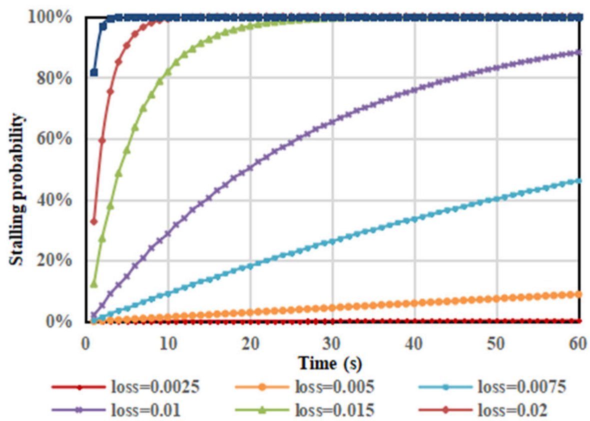

FIGURE 4. Probability of stalling in 30 FPS and 5ms re-transmission latency in simulation with 60M encoding rate.

is changed to be more than 0.75%, the probabilities of stalling are higher than 50% in 60s, resulting in an intolerable user experience for VR services.

In addition, simulations with a frame rate of 45 FPS are also conducted. As shown in the Figure 5, the re-transmission latency is set to be 2.5 ms. When the data packet loss is adjusted to be 0.25%, 0.5%, 0.75% and 1%, the analytical results of stalling probability are about 0.04%, 2.64%, 21.61% and 64%, respectively. When the data packet loss increases to be 1.5% and 2%, the probabilities of stalling increase more sharply to be almost 100%.

As illustrated in the Figure 6, The latency for each packet retransmission is changed to be 5ms. When the data packet loss is adjusted to be 0.1% and 0.25%, the simulated stalling probabilities are about 1.14% and 29.75%, respectively. When the data packet loss exceeds 0.5%, the probabilities of stalling increase close to the 100%. Compared with 30 FPS, the data packet loss of no stalling in 45 FPS is reduced by 0.4% and 0.9%. As the increase of FPS leads to more data transmitted in per second, and more packets loss are introduced into the transmission of network, however, the frame loss threshold for stalling in 30 FPS is similar with 45 FPS. Consequently, higher FPS is more likely to cause users to experience freezing and stalling.

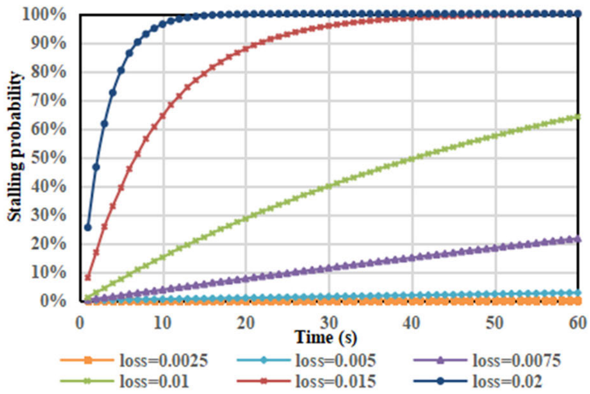

FIGURE 5. Probability of stalling in 45 FPS and 2.5 ms re-transmission latency in simulation with 60M encoding rate.

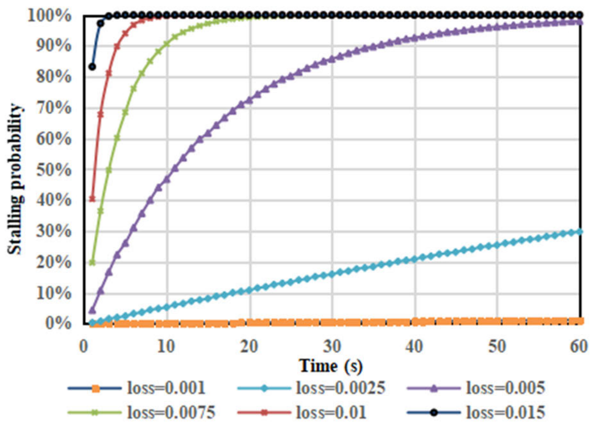

FIGURE 6. Probability of stalling in 45 FPS and 5 ms re-transmission latency in simulation with 60M encoding rate.

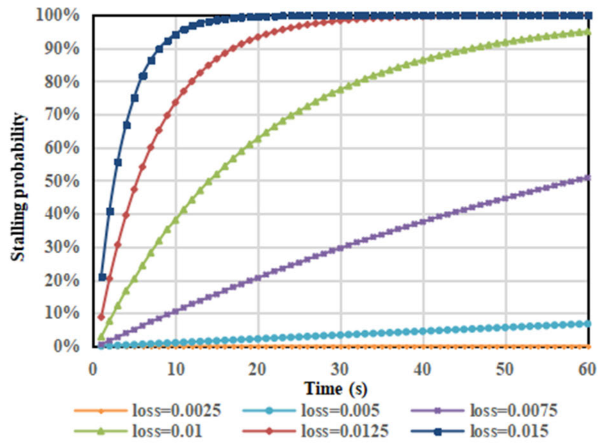

FIGURE 7. Probability of stalling in 60 FPS and 2.5 ms re-transmission latency in simulation with 60M encoding rate.

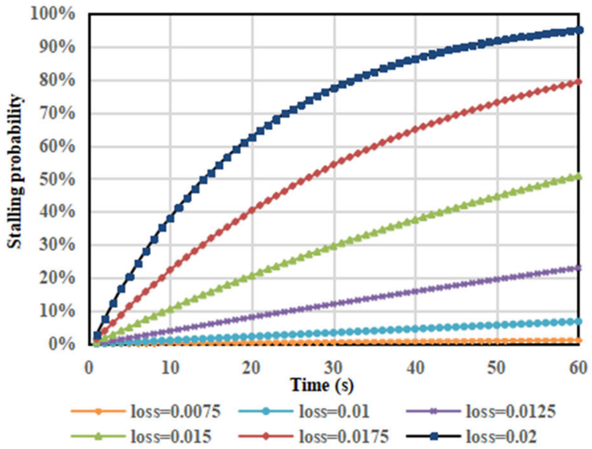

FIGURE 8. Probability of stalling in 60 FPS and 2.5 ms re-transmission latency in simulation with 30M encoding rate.

Finally, the XR service with 60 FPS is simulated with different encoding rate of 60 Mbps and 30 Mbps. In order to accelerate the convergence, some approximate methods and simplified calculations are introduced in the simulation program. As shown in the Figure 7, the re-transmission latency is set to be 2.5 ms, and the encoding rate is 60 Mbps. When the data packet loss is set to be 0.25%, 0.5%, 0.75% and 1%, the probabilities of stalling are about 0.06%, 6.74%, 50.8%, 95% in 60s, respectively. When the data packet loss is changed to be 1.25% and 1.5%, analytical results show that the stalling probability is nearly 100%.

As illustrated in the Figure 8, the encoding rate of XR service is changed to 30 Mbps in the 60 FPS. When the data packet loss is set to be 0.75%, 1%, 1.25%, 1.5% and 1.75%, the analytical results of stalling probability are about 1.10%, 6.74%, 22.96%, 50.84% and 79.31% in 60s, respectively. When data packet loss is higher than 2%, the probabilities of stalling are almost 100%. Compared with the different encoding rate under the same conditions of network performance, the higher encoding rate results in more data packet loss. The data packet loss of 1% is a typical network performance requirement, which is similar to as the service of voice over NR (VoNR) in mobile communication network. In the high-capacity scenarios and weak coverage scenarios of 5G commercial networks, the requirement of data packet loss of 1% is difficult to achieve.

In conclusion, the simulation results indicate that the network is supposed to provide the performance of data packet loss to be less than about 1.4%, which can ensure user to experience the cloud rendering VR service without feeling frozen or stalling. The FPS and encoding rate are the important factors affecting the stalling. With the increase of FPS and coding rate, the demand for data packet loss enhances accordingly. Additionally, re-transmission latency, the latency for each packet retransmission is also a significant factor, higher performance of re-transmission latency can reduce the requirements on data packet loss, but the capacity and throughput of network may be affected.

B. TEST RESULTS

In order to ensure the accurate of the simulation results, the experiments are conducted in the 5G network of commercial configuration at laboratory. As shown in Figure 9, the cloud rendering VR game named Archery Shooter is deployed on a HMD (PICO Neo3), and connected with a 5G user equipment (UE) via the universal serial bus (USB) for the

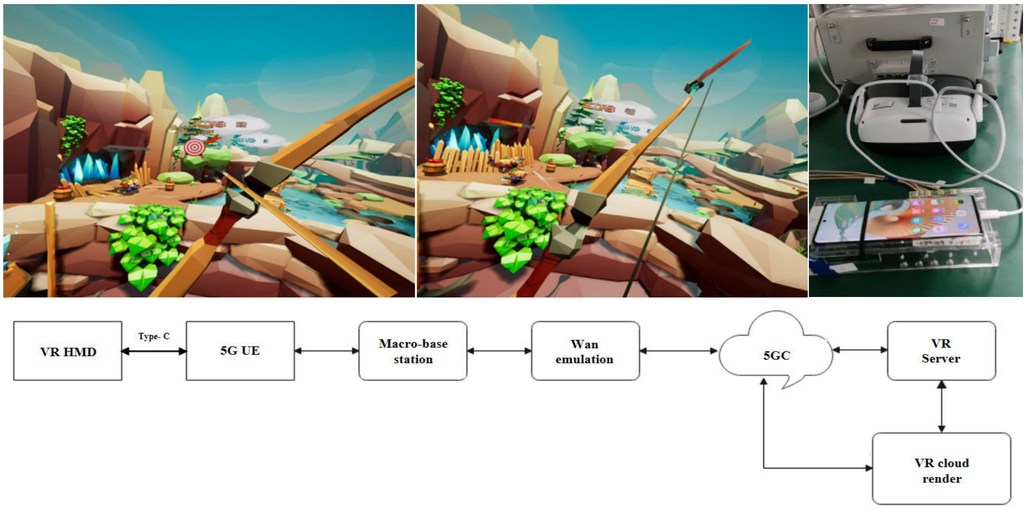

FIGURE 9. Schematic diagram of network architecture for experimental test and VR game.

TABLE 2. The results of simulation.

FPS	coding rate (Mbps)	Latency of re-transmission (ms)	Data packet loss of almost no stalling in 60s
60	60	2.5	0.5%
60	30	2.5	1%
45	60	2.5	0.5%
45	60	5	0.1%
30	60	2.5	1.4%
30	60	5	0.5%

data transmission. The 5G UE is connected into a 5G test cell, where a macro-base station with antenna number of 64T is applied. The bandwidth of cell is 100 MHz at the frequency of 4.9 GHz, which is a commercial 5G spectrum with the uplink/downlink configuration of the 3:7. The WAN emulation (Spirent Attero-100G) is deployed between the base station and 5GC. Based on a set data packet loss, the WAN emulation discards the data packet in periodic. Since data packets are transmitted in a random and unordered manner in the network [44], unpredictable packet loss patterns can be simulated in the experiment. The game server is deployed adjacent to the 5GC and the VR cloud game is rendered and encoded on a PC (CPU: i9-9900K, GPU: RTX 2080Ti, RAM: 32G). The resolution of the VR game is set to be 4K (4096 × 2160) at 60 FPS with the encoding rate of 30 and 60 Mbps. When the data packet loss is adjusted from 1% to 10% with step of 1%, the users experience the cloud rendering VR game at least 10 minutes per test, and subjectively record the number and duration of stalling. The network management platform measures the network quality indicators such as the data packet loss, transmission latency, throughput and other indicators.

As illustrated in the Figure 10, the cloud rendering VR game is configured with FPS of 60 and encoding output rate of 60 Mbps. The video stalling does not occur when the data packet loss is less than 1%. When the data packet loss is adjusted to be 2% and 3%, the average number of stalling is measured about 0.5 to 3.5 with the average stalling duration of 0.5 to 1s. The users subjectively report that the video frame is smooth and can tolerate a few slight stalling. When the data packet loss is set to be 4% and 5%, the average number of stalling is increased to be 7 to 10 times with stalling duration of 1 to 1.5s at 10 minute test. The user's subjective experiences often appeared unbearably stalling, the progress of the cloud rendering VR game can barely be maintained. When the data packet loss is changed to be 7%, the average number of stalling is measured about 50 in 10 minute test, which is subjectively intolerable to users with long time stalling and video frozen. The trend-line between the number of stalling and data packet loss can be measured to match as

$$\begin{aligned} n_s = & \, 271712 \cdot P(Packet\ loss|\Delta t)^3 \\ & - 12527 \cdot P(Packet\ loss|\Delta t)^2 \\ & + 250.37 \cdot P(Packet\ loss|\Delta t) - 0.7368 \end{aligned} \tag{17}$$

where the n_s is the number of stalling measured in the test.

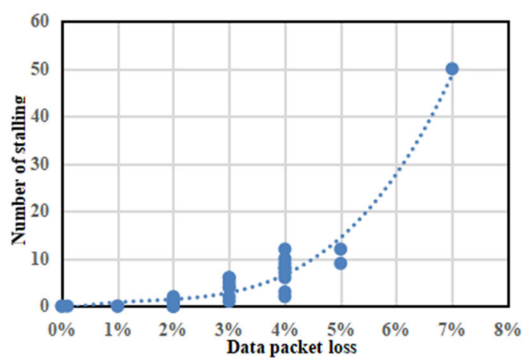

FIGURE 10. Number of the stalling in 60 FPS and 60 Mbps code rate in the test.

As shown in the Figure 11, the output rate of encoding is adjusted to be 30 Mbps. Comparing with the results of 60 Mbps, the data packet loss of no stalling is reduced to be 2% on the 10 minute test. When the data packet loss is changed to be 3% and 4%, the average number of stalling are measured to be about 3.25. When the data packet loss is adjusted to be more than 8%, the average number of stalling increases to be approximately 50. The trend-line between the number of stalling and data packet loss in 30 Mbps can be measured to match as

$$n_s = 50651 \cdot P(Packet\ loss|\Delta t)^3 + 1782.5 \cdot P(Packet\ loss|\Delta t)^2 - 44.24 \cdot P(Packet\ loss|\Delta t) - 0.0269 \quad (18)$$

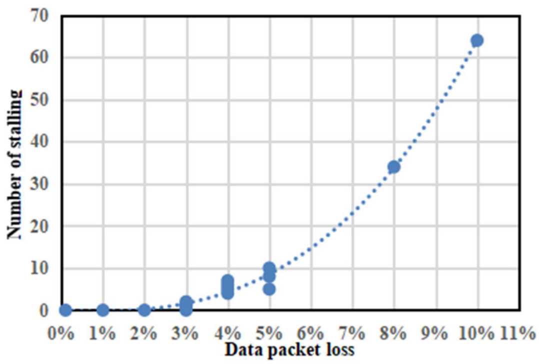

FIGURE 11. Number of the stalling in 60 FPS and 30 Mbps code rate in the test.

In the experimental test, the results show that network is supposed to provide the performance of data packet loss to be less than 1% and 2% in 60 FPS with 60 Mbps and 30 Mbps encoding rate, respectively. Comparing with the results of simulation, the data packet loss thresholds for no stalling are reduced by about 0.5% to 1%. Considering the individual differences in the tolerance of the video stalling, the count of stalling number and duration may involve statistical errors, the test result can prove that the proposed theoretical model and simulation are feasible and accurate.

Furthermore, in order to more comprehensively verify the capability of the proposed two-layer model in live network conditions, the experiments with different retransmission performance are also carried out. The FPS of cloud rendering VR gaming is set to be 60 and encoding rate is set to be 60 Mbps. Generally, the transmission modes for cloud rendering XR are set to be UM (Unacknowledged Mode) or AM (Acknowledged Mode) in the base station. In UM, only HARQ retransmission is available, and the retransmission latency is about 6-10 ms, depending on the retransmission threshold of HARQ [45]. In addition to HARQ retransmission, RLC (Radio Link Control) retransmission is used in AM [46]. When the data packet errors or loss occur in the RAN, the base station first initiates HARQ retransmissions, and if the transmission of the data packet is not completed within the predefined time period through HARQ retransmissions, the base station subsequently triggers RLC retransmissions. Typically, the retransmission latency of RLC is about 25ms. In the experiments, the users are placed in different locations which are near or far from the base station antenna. Various data packet loss is created by the different location, the transmission modes and the retransmission threshold of HARQ.

The experiment results are shown in the Table. 3. When the users are placed near from the base station antenna, the data packet loss is measured to be 2% both in AM and UM, the users report relatively good subjective experiences, with only a few slight stalling occurring in UM. When the users are placed far from the base station antenna and the network performance of retransmission is weak, the data packet loss is measured to be 7% and 9%. The average number of stalling are measured to be more than 4 and 14 times with the stalling duration of 2s, respectively. The subjective user experiences become intolerable and suffer from the sickness and dizziness. Since the wireless channel conditions are relatively stable in the experiments and transmission resources are sufficient, the retransmission of data packet loss may not significantly degrade conditions of no stalling. However, in the live network scenario with high capacity and weak coverage, the network may face insufficient resources to use high performance retransmissions, it is necessary to ensure the good experience for cloud rendering XR user by efficiently managing transmission resources. The proposed two-layer model can be used as a basic model for precise resource allocation in network transmission and contribute to the development of XR services in the future.

TABLE 3. The results of experiments with different retransmission performance.

The location	transmission modes	retransmission threshold of HARQ	Data packet loss	User's subjective experience
Near	AM	5	2%	No stalling
Near	UM	5	2%	A few slight stalling
Far	UM	5	7%	More 4 time stalling and duration time of 2s
Far	UM	4	9%	More 14 time stalling and duration time of 2s

V. CONCLUSION

In this research, the impact of network quality on video stalling for the cloud rendering XR is comprehensively analyzed and mathematically modeled. A frame-level data transmission framework with two-layer mapping model is proposed. The mathematical mapping relationship between network quality indicators and stalling is theoretically established, the probability of stalling can be predicted by various network performance conditions. The proposed model is simulated by Python, and simulation results indicate that data packet loss is supposed to be less than about 1.4%. In addition, the experimental tests are implemented to verify the accurate of the simulation in the 5G network of commercial configuration, the experimental results show that for the cloud rendering VR game with 60 Mbps and 30 Mbps encoding rate, the data packet loss thresholds for no stalling are measured to be less than 1% and 2%, respectively. According to the experimental results, the proposed theoretical model and simulation are feasible and accurate.

Moreover, the number and duration of stalling are also required to be considered in the model, and the mapping relation between network quality and other poor user experience of XR service are supposed to be revealed as well, such as black edge and artifact rate. That will be further discussed and studied in our future research. Furthermore, due to the individual difference, when the FPS of the video is momentarily below the threshold for smooth FPS in a short time, some individuals may not feel the stalling and video frozen in the experimental test. The proposed model of video stalling for cloud rendering XR could be further updated and optimized in future work.

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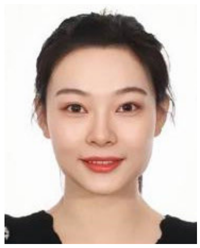

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